

Project ID :
25-26J-002

1. Topic (12 words max)

OceanSense: An IoT-Based Hybrid System for Real-Time Detection and Tracking of Marine Debris

2. Research group the project belongs to

CI - Computing Infrastructure

3. Specialization of the project belongs to

Computer Systems and Network Engineering(CSNE)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Marine debris, especially plastics, has reached crisis levels. **Eight million metric tons** of plastic enter the oceans annually, and an estimated 88% of ocean surface is littered^[1]. This pollution endangers wildlife and ecosystem health. Monitoring and cleaning such vast, dynamic environments is challenging. Existing approaches (satellites, manned patrols, or stationary sensors) are expensive or limited in coverage. Underwater sensing (acoustic/optical) is hampered by high attenuation and short range; above-surface drones and boats require human oversight or costly infrastructure.

There is a **gap for an affordable, scalable monitoring system** that can autonomously detect debris in real time over large areas. Key challenges include:

- **Limited Connectivity:** Marine environments lack reliable high-bandwidth links. Acoustic/UWOC links are short-range and energy-intensive, while satellites or cellular are not always available. LoRaWAN and other LPWANs offer long range with low power^[5], but at low data rates, requiring careful design of communication protocols.
- **Energy Constraints:** Buoyant nodes must run for extended periods with minimal maintenance. Swapping batteries in ocean is impractical. Thus, harvesting ambient energy (solar and wave) is essential to sustain operations^[8]. The system needs energy-aware scheduling to avoid node downtime.

- **Decentralized Coordination:** Relying on a central server or leader is risky (single point of failure, connectivity bottleneck). A **fully decentralized swarm** is needed so that nodes collaborate via local communication to maintain coverage, and optimize operations. Designing simple state-machine coordination based on neighbour messages (a practice in swarm robotics[3][2]) can achieve collective intelligence without complex overhead.
- **Edge AI on Constrained Devices:** Detecting debris requires computer vision. Cloud processing is infeasible given bandwidth limits; therefore, lightweight ML models must run on each node (Raspberry Pi 3B). Achieving acceptable accuracy and speed on such hardware is challenging (e.g. Tiny YOLO runs under 0.2 FPS on RPi 3[4])—necessitating model pruning and optimisation.

In summary, the problem is to develop a **decentralized, energy-efficient IoT network of mobile sensor nodes** that can perform on-device debris detection and adapt to environmental conditions. This addresses the urgent need for scalable, persistent ocean monitoring without relying on expensive centralised infrastructures.

References

[1] J. Li, K. Liu, and H. Zhang, "Global trends in marine plastic pollution and its ecological impacts," *IEEE Access*, vol. 10, pp. 55203–55216, 2022.

[2] [3] Decentralized construction of a global coordinate system in a large swarm of minimalistic robots | Swarm Intelligence | Springer Nature Link

<https://link.springer.com/article/10.1007/s11721-025-00251-4>

[4] A. Topouzelis, I. Papakonstantinou, and D. Garaba, "Detection of floating plastics from satellite and UAV imagery: Challenges and future directions," *Remote Sensing of Environment*, vol. 259, pp. 112414, 2021.

[3] Y. Luo, H. Wang, and C. Liu, "Energy-efficient design of underwater wireless sensor networks: A comprehensive survey," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 2, pp. 1025–1051, 2022.

[4] R. Diamant and L. Lampe, "Underwater acoustic communications: Challenges and future trends," *IEEE Journal of Oceanic Engineering*, vol. 42, no. 4, pp. 1447–1464, Oct. 2017.

[5] P. Casari and M. Zorzi, "Protocol design issues in underwater acoustic networks," *Computer Communications*, vol. 34, no. 17, pp. 2013–2025, Nov. 2011.

[8] Sustainable energy harvesting techniques for underwater aquatic systems with multi-source and low-energy solutions - ScienceDirect

<https://www.sciencedirect.com/science/article/abs/pii/S2210537925000460>

[6] (PDF) Aquatic Debris Monitoring & Detection using Raspberry Pi based 'AQUABOT'

https://www.researchgate.net/publication/318060173_Aquatic_Debris_Monitoring_Detection_using_Raspberry_Pi_based_'AQUABOT'

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed *OceanSense* system consists of **buoyant mobile sensor nodes** operating cooperatively. Each node is a **self-contained platform**

- **Hardware:** Each node mounts a Raspberry Pi, a compatible camera module, GPS receiver, IMU, environmental sensors, and an ultrasonic sensor for close-range obstacle detection. Propulsion is provided by small electric thrusters or a motor-driven rudder for basic maneuvering. A LoRa radio serves as the sole communication channel. Solar panels and a small wave-energy harvester power the node, connected to a rechargeable battery and power management circuits.
- **Software:** Onboard software is Linux-based. A lightweight state machine governs behavior with modes for exploring (patrolling), avoiding collisions, assisting others (converging on a detected debris cluster), and safe-return. Sensor data (GPS, IMU, environment) feeds into mode transitions. A compact neural network runs on the Pi to classify detected objects in camera frames as debris or not. Only the inference results (object class, confidence, location) are sent over LoRa, minimizing bandwidth. All nodes periodically broadcast their state and position via LoRa; neighbors use this to coordinate (avoid overlap, cover gaps) without central control.
- **Network & Coordination:** A custom LoR like MAC schedules transmissions by energy budget and priority (alerts highest). The decentralized protocol allows multi-hop relaying to extend range. Swarm-level objectives (coverage, redundancy) emerge from local rules: for instance, if a node detects missing neighbors or low network density, it may adjust its course or mode. Swarm behaviours leverage **finite-state-machine coordination**: each node's FSM changes in response to environmental cues and peer messages[\[3\]\[2\]](#).
- **Edge AI:** Debris detection is done entirely on-device. The choice of model (TinyML-optimised YOLO) balances accuracy and speed; existing reports show YOLOv4-tiny on RPi3 yields ~0.2 FPS[\[4\]](#), so optimisations (quantization, pruning) will be applied. This enables near real-time classification; detected events trigger alerts. By keeping raw image processing local and sending only metadata, we maintain responsiveness and save power.

This **decentralized** architecture contrasts with traditional centralised monitoring. No master node is needed – each robot is autonomous yet cooperative. LoRa enables **long-range links** with minimal power[\[5\]\[9\]](#), and the energy-harvesting powertrain aims for multi-year deployment. The solution leverages CSNE expertise in embedded hardware design, low-power wireless networking, and distributed algorithms to create a **real-time marine IoT swarm**.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The OceanSense project requires interdisciplinary expertise spanning embedded hardware, wireless communications, artificial intelligence, and marine data analysis. Each component of the system leverages specialized knowledge in order to ensure a resilient and intelligent IoT network for debris detection.

Embedded Hardware Design: Knowledge of buoyant electronics packaging and waterproofing (e.g. marine-grade enclosures) is required. Sensor integration (camera, GPS, IMU, ultrasonic) and actuator control (motors) demand experience with microcontrollers and interfaces. Understanding of solar panels and ocean wave generators is needed for the energy-harvesting subsystem.

Wireless IoT Communication: Proficiency with LPWAN/LoRa protocols and low-power MAC design is crucial[5][9]. This includes configuring LoRa modules (frequency, spreading factor), link budgeting in marine conditions, and network simulation tools like NS-3 (which has LoRaWAN modules) to test protocol performance. Knowledge of energy-aware scheduling (e.g. adaptive duty-cycling) will optimise battery life.

Distributed Systems & Swarm Robotics: The system requires designing and implementing decentralized algorithms. Key concepts include mesh and ad-hoc networks, distributed consensus, and failure recovery. Background in swarm robotics (finite-state machines, behavior-based coordination) is needed[3][2]. Familiarity with robotic frameworks (e.g. ROS, ARGoS simulator) will aid simulation and testing of multi-robot interactions.

Computer Vision & Edge AI: Expertise in lightweight neural networks is needed. This includes selecting or customising models (YOLO, MobileNet, etc.), dataset preparation (marine debris images), and quantization/pruning for embedded inference. Tools like TensorFlow Lite or PyTorch Mobile will be used. Understanding of performance metrics (FPS, accuracy) on constrained devices (Raspberry Pi 3B) is necessary; prior work shows RPi3 yields ~0.2–2 FPS for Tiny-YOLO[4].

Data Resources: We will use existing open datasets for marine debris to train/test the model. GPS waypoint and oceanographic data (currents, visibility) may be used to simulate realistic conditions. Calibration requires ground-truth debris scenarios.

8. Objectives and Novelty

Main Objective: To develop a decentralized, energy-efficient swarm IoT system with edge-based machine learning for real-time marine debris detection and adaptive coordination using LoRa communication.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
JAYASAK D L C IT22908360	Node Design & Self-Healing Network Develop a low-cost, buoyant IoT node capable of reliable operation in marine environments.	Design and buoyant sensor nodes with cameras, sonar, ultrasonic. Integrate LoRa radios. Implement solar + wave energy harvesting modules.	Unlike centralized recovery schemes, OceanSense uses a lightweight self-healing protocol : each node autonomously detects loss of neighbors and re-establishes connectivity through adaptive retransmission or route changes. This energy-efficient fault tolerance avoids complex global coordination.
RATHNAYAKA G R M S D IT22228130	Energy-Aware Adaptive Communication MAC Protocol designing Design an adaptive protocol stack that maximizes network lifetime and resilience.	Design a MAC protocol that schedules transmissions based on energy forecasts (solar/wave). Implement swarm-assisted redundancy for fault tolerance. Simulate protocols with NS-3 UAN and validate resilience metrics.	Novel energy-aware MAC with dual power harvesting prediction , coupling dual energy harvesting forecasts with QoS scheduling and fault-tolerant mobility. Unlike conventional MACs, this design exploits swarm adaptability to extend lifetime and ensure continuous coverage.
ABEYSINGHE P T G S R IT22253576	Environment-Aware Swarm Coordination Enable autonomous swarm behavior for efficient debris	Develop swarm mobility algorithms for repositioning and area coverage. Create heuristic rules for swarm-based redundancy. Implement	Applies environment-based coordination in a swarm context. Nodes adapt their roles and coverage dynamically to real-time conditions. This adaptive swarm

	coverage.	decentralized coverage strategy: nodes share location and use simple rules to fill gaps or avoid overlap.	behavior improves efficiency and resilience in variable marine settings.
DISSANAYAKA D M T N IT22169730	Edge ML for Debris Detection & Drift Prediction Deploy lightweight ML pipelines on nodes for autonomous debris detection and forecasting with a central ML model.	Assemble or curate a dataset of annotated marine debris images.train a lightweight CNN (YOLOv8n) for debris detection; perform model compression for Raspberry Pi deployment. Implement on-device inference pipeline: capture image, downscale, run CNN, and output classification with confidence. Transmit only metadata (type, confidence, location).	Deploying deep learning at the edge on resource-constrained buoys for environmental monitoring. The innovation is in efficient on-device debris identification, minimizing communications.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
JAYASAK D L C IT22908360	IoT Sensor Node Design & Self-Healing Network Self-Healing Hybrid Sensor Node: This component covers the hardware design of each buoyant node. The student will select and integrate components (Pi , camera, sensors, actuators) and design the power system (solar + wave harvesters). Expertise in embedded hardware and C/C++/Python programming is required). The inclusion of swarm-assisted self-healing mechanisms connects networking with system recovery strategies, directly reflecting CSNE expertise in resilient network design.
RATHNAYAKA G R M S D IT22228130	Energy-Aware MAC protocol This component focuses on designing cross-layer protocols that integrate energy harvesting with adaptive MAC scheduling, a core area of network engineering. It aligns with CSNE through fault-tolerant distributed systems, resilience in IoT networks, and advanced simulation/validation (NS-3 UAN, Aqua-Sim Next). The inclusion directly reflecting CSNE expertise in resilient network design.
ABEYSINGHE P T G S R IT22253576	Adaptive Swarm Coordination This component will program the nodes to share local information and make collective decisions. For instance, nodes could broadcast wave height or node density and adjust their behavior (slower movement in rough seas, or re-spacing in calm conditions). Simple coverage algorithms (e.g. potential fields or leader election for gap filling) will be used. Expertise in robotics or distributed computing is important. The novelty is using environmental context to steer swarm behavior embedding real-time adaptation into the coordination logic, enhancing mission effectiveness without added hardware. This reflects CSNE expertise in communication efficiency, distributed algorithms, and scalable IoT intelligence.
DISSANAYAKA D M T N IT22169730	Edge Machine Learning The student will select a suitable lightweight model (like YOLOv8n), train it on a marine debris dataset, and optimize it for the Pi. They must manage the end-to-end pipeline: image capture, preprocessing, inference, and result transmission. Tools like OpenCV will be used. The innovation is on-device debris recognition on a moving platform. This component demonstrates edge AI on minimal hardware, reducing data transmission by sending only detection results. This reflects CSNE expertise in communication efficiency, distributed machine learning, and scalable IoT intelligence.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Dinithi	Pandithage	
Co-Supervisor	Ms.	Narmada	Gamage	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.